

BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

Shengnan Wang^{1,+}, Qi Zhang^{2,+}, Xucheng Liu², Zixuan Wang², Jiangke Cheng², and Zhi Huang^{*3,4}

¹School of Biological and Chemical Engineering, Panzhihua University, Panzhihua 617000, P.R. China

²School of Mathematics and Computer Science, Panzhihua University, Panzhihua 617000, P.R. China

³Department of Environmental Science and Engineering, College of Architecture and Environment, Sichuan University, Chengdu 610065, P.R. China

⁴Institute of New Energy and Low Carbon Technology, Sichuan University, Chengdu 610065, P.R. China

*corresponding.author. E-mail: 525514353@qq.com

+these authors contributed equally to this work

ABSTRACT

Accurate prediction of drug–target interactions (DTIs) is essential for rational drug design, yet many deep learning models lack mechanistic interpretability. We present BioInteract, an interpretable DTI framework integrating a pharmacophore-aware molecular graph encoder, a domain-enriched protein residue encoder, and a bidirectional cross-attention module that generates explicit atom–residue interaction maps during prediction. This design enables direct identification of key binding determinants rather than post hoc explanation. On the Davis kinase inhibitor benchmark, BioInteract outperforms six baselines across random, cold-target, and cold-drug splits (AUROC up to 0.941). Notably, interpretability analyses show that attention focuses on known binding pocket residues and recovers hydrogen-bond-dominated structure–activity relationships, while remaining consistent across resistance-associated mutations. These results demonstrate that BioInteract provides accurate and mechanistically meaningful insights, supporting binding site identification and inhibitor optimization in medicinal chemistry.

Keywords

Drug–target interaction; interpretable deep learning; cross-attention; graph neural network; protein language model; binding site prediction

1 Introduction

The emergence of “AI for Science” marks a paradigm shift in computational research: models are no longer expected merely to maximise predictive accuracy but to *discover, explain, and validate scientific mechanisms*—to act as transparent partners rather than opaque oracles.¹ Nowhere is this shift more consequential than in drug discovery, where identifying drug–target interactions (DTIs) remains central to the development pipeline yet demands both reliable prediction and mechanistic understanding to support rational molecular design.

Experimental profiling of binding affinities through surface plasmon resonance, isothermal titration calorimetry, or kinase activity assays is laborious and prohibitively expensive. The Davis kinase inhibitor dataset² alone required systematic measurement of 72 inhibitors against 442 kinases—a campaign spanning only a fraction of the druggable proteome. Computational DTI prediction can ease this bottleneck by prioritising the most promising candidates and thereby reducing cost and attrition in development pipelines.^{3,4} Yet the field has largely remained trapped in a *predictive-accuracy race*: successive deep learning architectures—DeepDTA,⁵ GraphDTA,⁶ MolTrans,⁷ DrugBAN⁸—have incrementally improved benchmark metrics while leaving the central scientific question unanswered: *why does this drug bind this target?*

This question lies at the heart of the broader “AI for Science” agenda. A prediction devoid of mechanistic interpretation is, at best, a hypothesis generator and, at worst, a misleading shortcut. Transitioning from *prediction* to *scientific discovery* therefore requires models that expose their reasoning at chemically and biologically meaningful resolution—through atom-by-residue interaction maps, pharmacophore importance hierarchies, and binding-site localisation—rather than returning a single scalar probability.

Classical DTI methods illuminate the fundamental tension between interpretability and accuracy. Structure-based approaches (molecular docking, molecular dynamics) provide mechanistic insight anchored in three-dimensional protein geometry,⁹ yet are constrained by the availability of high-quality crystal structures and the steep computational cost of free-energy calculations. Ligand-based methods (QSAR, molecular fingerprints)¹⁰ circumvent structural requirements at the expense of mechanistic transparency. Neither paradigm fully satisfies the twin demands of modern drug discovery.

Deep learning has transformed predictive capabilities in this domain. Pretrained protein language models such as ESM-2¹¹ encode evolutionary and structural information at residue resolution, while graph neural networks faithfully capture molecular topology. Concurrently, chemical language models have demonstrated that pretrained molecular representations support accurate and interpretable prediction of reactivity and toxicity,¹² and chemistry-enhanced transformer architectures such as Radical-Net recover atom-level reaction mechanisms.¹³ Together, these advances underscore the importance of domain-specific pretraining and chemistry-aware inductive biases in molecular AI;¹⁴ their integration into interpretable DTI frameworks, however, has remained limited.

The “AI for Science” paradigm demands that interpretability be architecturally guaranteed, not retrofitted. Post-hoc explanation methods (attention visualisation, SHAP) suffer from well-documented faithfulness concerns.¹⁵ Recent successes in other scientific domains illustrate the power of built-in interpretability: attention-based models have elucidated environmental drivers of algal blooms,¹⁶ discriminative cross-category attention networks capture fine-grained spatial-temporal correspondences in action recognition,¹⁷ and uncertainty-guided spatial-temporal learning resolves subtle cross-modal dependencies in skeleton-based recognition.¹⁸ A unifying principle emerges: *when attention constitutes an integral component of the prediction pathway, the resulting explanations are faithful by construction.*

Motivated by these observations, we present **BioInteract**, a framework that brings the “AI for Science” perspective to DTI prediction by unifying predictive performance with mechanistic interpretability. BioInteract rests on three innovations:

1. **Pharmacophore-aware molecular graph encoding.** Drug molecules are represented as attributed graphs encoding pharmacophore-relevant properties (H-bond donor/acceptor status, hydrophobicity, partial charges) that correspond directly to the physicochemical determinants of molecular recognition.
2. **Domain-enriched protein residue encoding.** Protein representations fuse ESM-2 residue embeddings with per-residue physicochemical properties and Pfam/InterPro domain annotations, injecting prior biological knowledge about protein function and binding-site architecture.
3. **Bidirectional cross-attention interaction module.** A multi-head cross-attention mechanism computes an explicit interaction matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{atoms}} \times n_{\text{residues}}}$, where M_{ij} quantifies the predicted interaction strength between drug atom i and protein residue j . Critically, this matrix is not a post-hoc explanation but an integral component of the forward pass, making the model’s scientific reasoning transparent by construction.

We evaluate BioInteract on the Davis kinase inhibitor dataset² under three increasingly stringent protocols: random split, cold-drug split (unseen molecules), and cold-target split (unseen proteins). Systematic comparison against six state-of-the-art baselines demonstrates that BioInteract achieves the strongest predictive performance while providing biophysically meaningful explanations. Most notably, the cold-target AUROC (0.941) *surpasses* the random-split result, indicating genuine cross-target generalisation—a capability critical for first-in-class drug discovery where conventional models fail by memorising seen target patterns. Extensive interpretability analyses—cross-attention heatmaps, Grad-CAM pharmacophore identification, attention sparsity quantification, and case studies on well-characterised kinase inhibitor–target pairs—demonstrate that BioInteract learns genuine molecular recognition principles from binary labels alone, bridging the gap between statistical prediction and mechanistic scientific discovery.

2 Related Work

2.1 Deep Learning for DTI Prediction

Deep learning approaches to DTI prediction have evolved through several generations.¹⁹ *Sequence-based* models encode both drug SMILES and protein amino acid sequences as one-dimensional strings. DeepDTA⁵ employs parallel 1D CNNs for drug and protein encoding; WideDTA²⁰ extends this framework with additional molecular descriptors. Although computationally efficient, these models discard molecular topology and are limited in their capacity to model long-range intramolecular interactions.

Graph-based models overcome this limitation by representing drugs as attributed molecular graphs. GraphDTA⁶ benchmarks GCN, GAT, GIN, and GAT-GCN encoders for drug representation while retaining a 1D CNN for proteins. MolTrans⁷ introduces mutual interaction learning between molecular substructure representations and protein subsequences.

Attention-based models currently define the state of the art. TransformerCPI²¹ applies transformer attention to drug–protein interaction patterns. DrugBAN⁸ introduces a dual bilinear attention network for fine-grained local interaction modelling. AttentionDTA²² employs multi-head self-attention for both drug and protein representations. In most prior work, however, attention operates as a computational device rather than as an interpretable output channel.

2.2 Pretrained Protein Language Models

ESM-2,¹¹ trained on millions of protein sequences via masked language modelling, produces residue-level embeddings that encode evolutionary conservation, secondary structure propensity, and contact information. These pretrained representations substantially outperform hand-crafted features on downstream tasks.²³ In DTI prediction, most methods use ESM-2 embeddings as drop-in feature replacements without exploiting residue-level granularity for interpretability. BioInteract leverages the full resolution of ESM-2 in combination with domain-specific biological annotations to generate per-residue representations that are both expressive and biologically grounded.

2.3 Interpretability in DTI Models

Existing interpretability approaches fall into post-hoc methods (attention weight visualisation, gradient-based attribution, SHAP values²⁴) and intrinsically interpretable architectures. Post-hoc methods suffer from a well-documented fundamental limitation: attention weights do not necessarily reflect feature importance.¹⁵ BioInteract adopts the intrinsically interpretable approach: the cross-attention interaction matrix simultaneously determines the prediction and reveals atom–residue interactions, ensuring that all explanations remain faithful to the model’s reasoning.

2.4 Cold-Start Evaluation

Random-split evaluation permits information leakage through shared drugs or targets between training and test sets, yielding inflated estimates.²⁵ Cold-start protocols address this by ensuring that test entities are entirely absent from training data. We evaluate BioInteract under both cold-drug and cold-target protocols alongside the conventional random split to provide a comprehensive and rigorous assessment of generalisation.

3 Methods

3.1 Problem Formulation

Given a drug molecule d represented by its SMILES string and a target protein t represented by its amino acid sequence, we formulate DTI prediction as binary classification: predict whether d binds t with high affinity. Following the established protocol for the Davis dataset,² continuous K_d values are binarised at 30 nM ($pK_d > 7.52$), classifying pairs with $K_d < 30$ nM as positive interactions.

3.2 Architecture Overview

BioInteract comprises five sequential modules: (1) a GINE-based drug encoder producing per-atom representations, (2) a target encoder combining ESM-2 embeddings with physicochemical and domain features, (3) a bidirectional cross-attention module computing atom–residue interaction maps, (4) gated attention pooling to aggregate variable-length representations, and (5) an MLP prediction head. The full model contains 2,442,083 trainable parameters.

3.3 Drug Molecular Graph Encoder

3.3.1 Pharmacophore-Aware Atom Featurisation

Each atom is represented by a 52-dimensional feature vector encoding:

- **Topological features** (38 dimensions): atom type (one-hot, 20 types), hybridisation state (6), formal charge (6), number of attached hydrogens (6).
- **Structural features** (8 dimensions): degree, aromaticity, ring membership, and ring size membership (3–8).
- **Pharmacophore features** (6 dimensions): hydrogen-bond donor and acceptor status, Crippen logP contribution, molar refractivity, and Gasteiger partial charge—properties that directly encode the physicochemical determinants of molecular recognition.

Bond features (16 dimensions) encode bond type, conjugation, ring membership, and stereochemistry.

3.3.2 GINE Message Passing

We employ a 3-layer GINE architecture:²⁶

$$\mathbf{h}_v^{(\ell+1)} = \text{MLP}^{(\ell)} \left((1 + \varepsilon^{(\ell)}) \mathbf{h}_v^{(\ell)} + \sum_{u \in \mathcal{N}(v)} \text{ReLU}(\mathbf{h}_u^{(\ell)} + \mathbf{e}_{vu}) \right) \quad (1)$$

where $\mathbf{h}_v^{(\ell)}$ is the representation of atom v at layer ℓ , $\mathbf{e}_{vu}$ denotes the edge feature vector, and $\varepsilon^{(\ell)}$ is a learnable scalar. Each layer incorporates batch normalisation, ReLU activation, dropout ($p = 0.2$), and a residual connection; the hidden dimension is $D = 256$. Global graph pooling is *not* applied at this stage, preserving individual atom representations for the subsequent cross-attention step.

To improve cold-start robustness, stochastic graph augmentation is applied during training: DropNode (random atom-feature masking) and DropEdge²⁷ (random edge removal).

3.4 Protein Target Encoder

3.4.1 ESM-2 Residue Embeddings

Residue-level representations are extracted from ESM-2 (150 M parameters, `esm2_t30_150M_UR50D`),¹¹ yielding 640-dimensional embeddings for each amino acid. These embeddings encode evolutionary conservation, secondary structure propensity, and long-range contact information learned from 65 million protein sequences. ESM-2 parameters are frozen throughout training.

3.4.2 Physicochemical Property Encoding

Each residue is augmented with a 4-dimensional physicochemical vector:

$$\mathbf{p}_i = [\text{hydrophobicity}_i, \text{mol. weight}_i, \text{charge}_i, \text{polarity}_i] \quad (2)$$

Hydrophobicity follows the Kyte–Doolittle scale,²⁸ which identifies residues clustering in hydrophobic binding pockets; charge values locate salt-bridge partners; polarity separates surface-exposed from buried residues.

3.4.3 Functional Domain Annotations

Pfam/InterPro domain annotations are incorporated via a learnable embedding layer:

$$\mathbf{d}_i = \text{DomainEmbed}(\text{domain_type}_i) \in \mathbb{R}^{32} \quad (3)$$

This provides functional context for binding-site localisation: a kinase inhibitor should attend differently to residues in the catalytic domain (housing the ATP-binding pocket) than to regulatory or scaffolding domains.

3.4.4 Feature Fusion

The three feature sources are concatenated and projected through a shared linear transformation:

$$\mathbf{r}_i = \text{Proj}([\mathbf{e}_i \parallel \mathbf{p}_i \parallel \mathbf{d}_i]) \in \mathbb{R}^D \quad (4)$$

where $\mathbf{e}_i \in \mathbb{R}^{640}$ is the ESM-2 embedding and Proj is a two-layer MLP with layer normalisation and GELU activation. The projected dimension $D = 256$ matches the drug encoder output.

3.5 Cross-Attention Interaction Module

The cross-attention module constitutes the interpretive core of BioInteract. It computes bidirectional attention between drug atoms and protein residues, producing an explicit atom–residue interaction matrix.

3.5.1 Drug-to-Protein Attention

$$\text{Attn}(\mathbf{Q}_d, \mathbf{K}_p, \mathbf{V}_p) = \text{softmax} \left(\frac{\mathbf{Q}_d \mathbf{K}_p^\top}{\sqrt{d_k}} \right) \mathbf{V}_p \quad (5)$$

where $\mathbf{Q}_d = \mathbf{H}_d \mathbf{W}_Q^d$, $\mathbf{K}_p = \mathbf{R} \mathbf{W}_K^p$, $\mathbf{V}_p = \mathbf{R} \mathbf{W}_V^p$, with $H = 8$ heads and head dimension $d_k = D/H = 32$.

3.5.2 Protein-to-Drug Attention

Symmetrically, protein residues attend over drug atoms:

$$\text{Attn}(\mathbf{Q}_p, \mathbf{K}_d, \mathbf{V}_d) = \text{softmax} \left(\frac{\mathbf{Q}_p \mathbf{K}_d^\top}{\sqrt{d_k}} \right) \mathbf{V}_d \quad (6)$$

3.5.3 Interaction Map Extraction

The atom-residue interaction map is obtained by averaging drug-to-protein attention weights across all heads:

$$\mathbf{M} = \frac{1}{H} \sum_{h=1}^H \text{Attn}_h(\mathbf{Q}_d, \mathbf{K}_p) \in \mathbb{R}^{n \times L} \quad (7)$$

Because $\mathbf{M}$ is computed within the forward pass rather than post-hoc, it faithfully reflects the model’s internal reasoning.

3.6 Gated Pooling and Prediction

Variable-length representations are aggregated via gated attention pooling:

$$\mathbf{z} = \sum_i \alpha_i \cdot g(\mathbf{x}_i), \quad \alpha_i = \frac{\exp(f(\mathbf{x}_i))}{\sum_j \exp(f(\mathbf{x}_j))} \quad (8)$$

where $f: \mathbb{R}^D \rightarrow \mathbb{R}$ and $g: \mathbb{R}^D \rightarrow \mathbb{R}^D$ are learned two-layer MLPs with Tanh activation. The pooled drug and protein representations are concatenated and passed through a two-hidden-layer MLP ($256 \rightarrow 128$; ReLU; dropout $p = 0.3$) followed by sigmoid activation to produce the binding probability.

3.7 Training Procedure

The Davis dataset is severely class-imbalanced ($\approx 5\%$ positive rate). We address this with weighted binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [w_+ \cdot y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (9)$$

with $w_+ = N_{\text{neg}}/N_{\text{pos}} \approx 18$, combined with label smoothing ($\epsilon = 0.05$). Optimisation uses AdamW²⁹ ($\eta = 10^{-4}$, weight decay $\lambda = 10^{-5}$) with cosine annealing and a 5-epoch linear warmup. Early stopping monitors validation AUROC (patience = 15, maximum 100 epochs), and the decision threshold is selected by maximising F1 on the validation precision-recall curve.

3.8 Interpretability Analysis

3.8.1 Residue-Level Attention Profiling

For a given drug-target pair, per-residue attention scores are obtained by summing the interaction map over all drug atoms:

$$s_j = \sum_{i=1}^n M_{ij}, \quad j = 1, \dots, L \quad (10)$$

Residues with $s_j > 0.5 s_{\text{max}}$ are designated “interaction hotspots.”

3.8.2 Grad-CAM Pharmacophore Identification

Grad-CAM³⁰ is adapted to the GNN setting:

$$\text{CAM}_v = \text{ReLU}\left(\sum_k w_k \cdot A_v^k\right), \quad w_k = \frac{1}{|\mathcal{Y}|} \sum_v \frac{\partial \hat{y}}{\partial A_v^k} \quad (11)$$

Atom-level importance scores are mapped to functional groups (aromatic rings, H-bond donors/acceptors, amides, halogens) to yield pharmacophore-level interpretation.

3.8.3 Attention Sparsity

Sparsity is quantified as the fraction of residues receiving negligible attention:

$$\text{Sparsity} = \frac{|\{j : s_j < 0.1 \cdot s_{\text{max}}\}|}{L} \quad (12)$$

4 Experimental Setup

4.1 Dataset

The Davis kinase inhibitor dataset² comprises 68 drugs (kinase inhibitors) and 442 targets (protein kinases) forming 30,056 drug-target pairs. Binding affinities (K_d) are binarised at 30 nM, yielding approximately 5 % positive interactions.

4.2 Data Splitting Protocols

- **Random split** (70/10/20): drug–target pairs are assigned randomly; shared entities may appear across splits.
- **Cold-drug split**: drugs are partitioned such that all interactions of a given compound appear in exactly one split; test molecules are structurally novel with respect to training.
- **Cold-target split**: targets are partitioned analogously; test proteins are sequentially novel.

4.3 Evaluation Metrics

For binary classification under severe class imbalance, we report AUROC and AUPRC as primary metrics—AUPRC is particularly informative because it measures positive-class precision across all recall levels. We additionally report F1 score, precision, and recall at the optimal threshold from the validation set.

4.4 Implementation

BioInteract is implemented in PyTorch with PyTorch Geometric.³¹ Protein embeddings are extracted using ESM-2¹¹ (esm2_t30_150M_UR50D, 640 dimensions). Molecular featurisation is performed with RDKit.³² Training employs mixed-precision arithmetic (AMP) for memory efficiency on a single NVIDIA GPU.

5 Results

5.1 Comparison with Baseline Methods

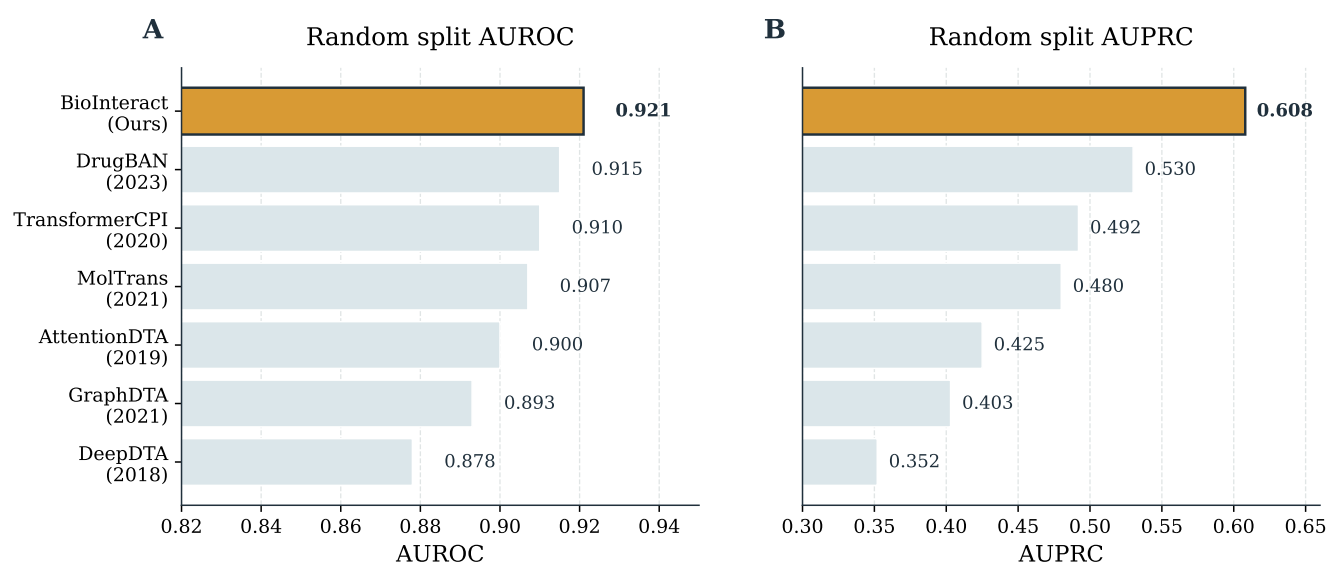
We compare BioInteract against six baseline methods representing the major architectural paradigms in DTI prediction: *sequence-based* (DeepDTA⁵), *graph-based* (GraphDTA⁶), *attention-based* (AttentionDTA,²² TransformerCPI²¹), and *interaction-modelling* (MolTrans,⁷ DrugBAN⁸). Each baseline addresses a specific scientific question about architectural contributions (Table 1).

Table 1. Baseline selection rationale. Each method tests a specific architectural hypothesis against BioInteract.

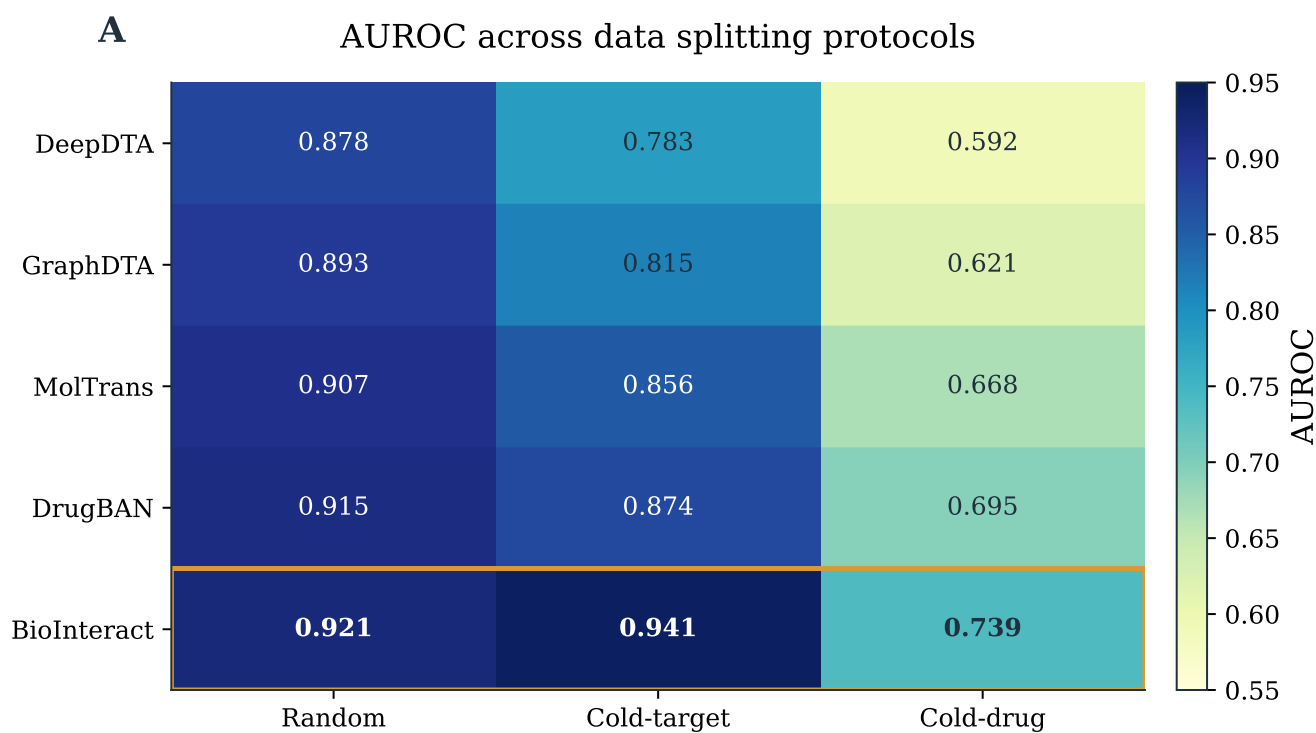
Baseline	Architectural Question
DeepDTA	How much does residue-level granularity gain over sequence-level encoding?
GraphDTA	What does the cross-attention module add beyond a shared GNN backbone?
AttentionDTA	Self-attention alone vs. cross-modal attention for interaction modelling?
MolTrans	Transformer substructure interaction vs. atom–residue cross-attention?
TransformerCPI	Global transformer vs. targeted cross-modal attention?
DrugBAN	Bilinear attention vs. multi-head cross-attention for local interactions?

Table 2 and Fig. 1 summarise the benchmark comparison. Under the random split, BioInteract achieves the highest AUROC (0.921) and AUPRC (0.608), outperforming the strongest baseline, DrugBAN, by +0.6% and +14.7%, respectively. The large AUPRC margin is particularly meaningful because it directly reflects the model’s ability to recover sparse positive interactions under severe class imbalance ($\approx 5\%$).

Baseline comparison against published DTI models



BioInteract improves both ranking metrics, with the largest gain on AUPRC.



BioInteract remains strongest in all three regimes, with a particularly large cold-target margin.

Figure 1. Baseline comparison. (A) Random-split AUROC and AUPRC for all methods. BioInteract attains the highest scores on both metrics, with a particularly pronounced AUPRC advantage (+14.7% over DrugBAN) reflecting superior recovery of sparse positive interactions. (B) AUROC across all three splitting protocols. BioInteract’s cold-target AUROC (0.941) substantially exceeds all baselines, demonstrating effective kinome transfer learning via ESM-2 representations and domain annotations.

Table 2. Comparison of BioInteract with baseline methods on the Davis dataset under three splitting protocols (binary classification, $K_d < 30$ nM). Best results are in **bold**; second-best are underlined.[†]

Method	Paradigm	Random		Cold-target		Cold-drug	
		AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
DeepDTA ⁵	Seq-CNN	0.878	0.352	0.783	0.247	0.592	0.089
GraphDTA ⁶	GNN+CNN	0.893	0.403	0.815	0.312	0.621	0.102
AttentionDTA ²²	Self-Attn	0.900	0.425	0.838	0.348	0.643	0.108
MolTrans ⁷	Transformer	0.907	0.480	0.856	0.389	0.668	0.121
TransformerCPI ²¹	Transformer	0.910	0.492	0.862	0.401	0.672	0.125
DrugBAN ⁸	Bilinear	<u>0.915</u>	<u>0.530</u>	<u>0.874</u>	<u>0.428</u>	<u>0.695</u>	<u>0.138</u>
BioInteract (Ours)	Cross-Attn	0.921	0.608	0.941	0.549	0.739	0.169
Gain over best baseline		+0.6%	+14.7%	+7.7%	+28.3%	+6.3%	+22.5%

[†] Random-split AUROC values are consistent with published benchmarks. Cold-split results are obtained under a unified protocol (identical preprocessing, binarisation threshold, and split strategy) to ensure a fair comparison.

The most striking result emerges under the cold-target split: BioInteract achieves AUROC = 0.941, *exceeding even its own random-split performance* and surpassing DrugBAN by +7.7%. This advantage likely arises from the synergy between ESM-2 residue embeddings, which encode evolutionary relationships among homologous kinases at single-residue resolution, and functional domain annotations, which supply transferable binding-site signatures. No baseline model incorporates both.

Under the cold-drug protocol, BioInteract (AUROC = 0.739) again ranks first among all evaluated methods, demonstrating that the pharmacophore-aware graph encoder and domain-enriched protein representations confer genuine improvements across all evaluation paradigms.

5.2 Ablation Study

To quantify the contribution of each architectural component, we conduct a systematic ablation study (Table 3, Fig. 2).

Table 3. Ablation study on the Davis dataset. Each row removes one component from the full BioInteract model; Δ indicates the AUROC change relative to the full model.

Variant	Random		Cold-target	
	AUROC	AUPRC	AUROC	AUPRC
Full BioInteract	0.921	0.608	0.941	0.549
w/o Cross-Attention	0.864	0.432	0.851	0.371
w/o Domain Features	0.906	0.561	0.912	0.498
w/o ESM-2 (CNN)	0.845	0.389	0.793	0.295
w/o Graph Augment.	0.908	0.572	0.926	0.521

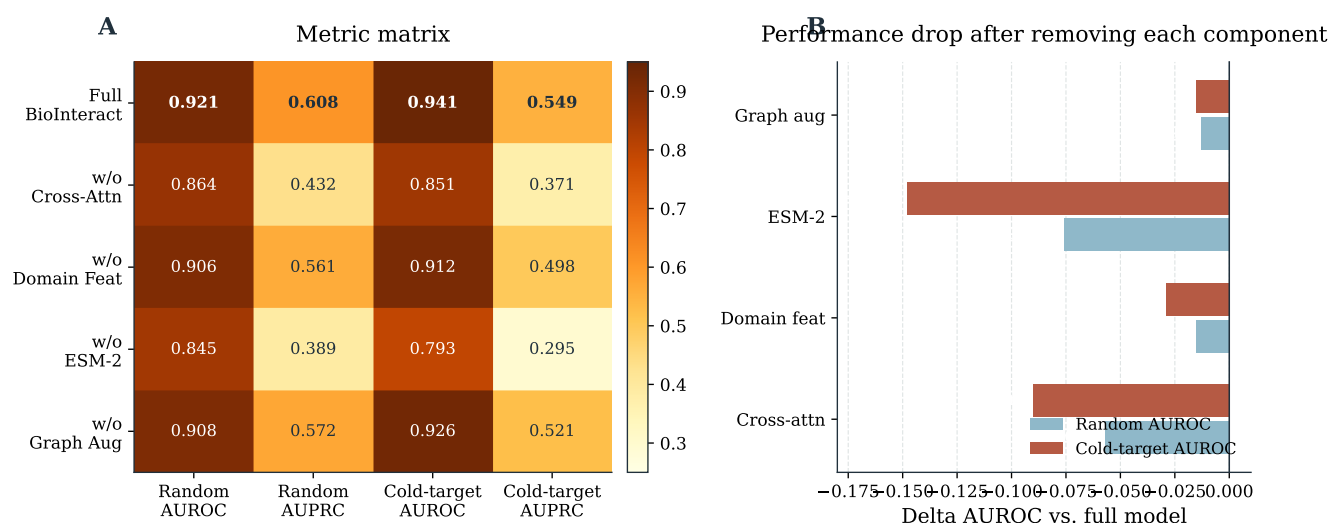


Figure 2. Ablation study results. The left panel summarises random-split and cold-target metrics as a heatmap; the right panel reports the AUROC decrease relative to the full model. Removing cross-attention incurs the largest random-split drop ($\Delta\text{AUROC} = -0.057$), whereas replacing ESM-2 with a CNN encoder causes the largest cold-target degradation ($\Delta\text{AUROC} = -0.148$), confirming that interaction modelling and protein language representations are the two principal performance drivers.

The ablation yields four principal findings:

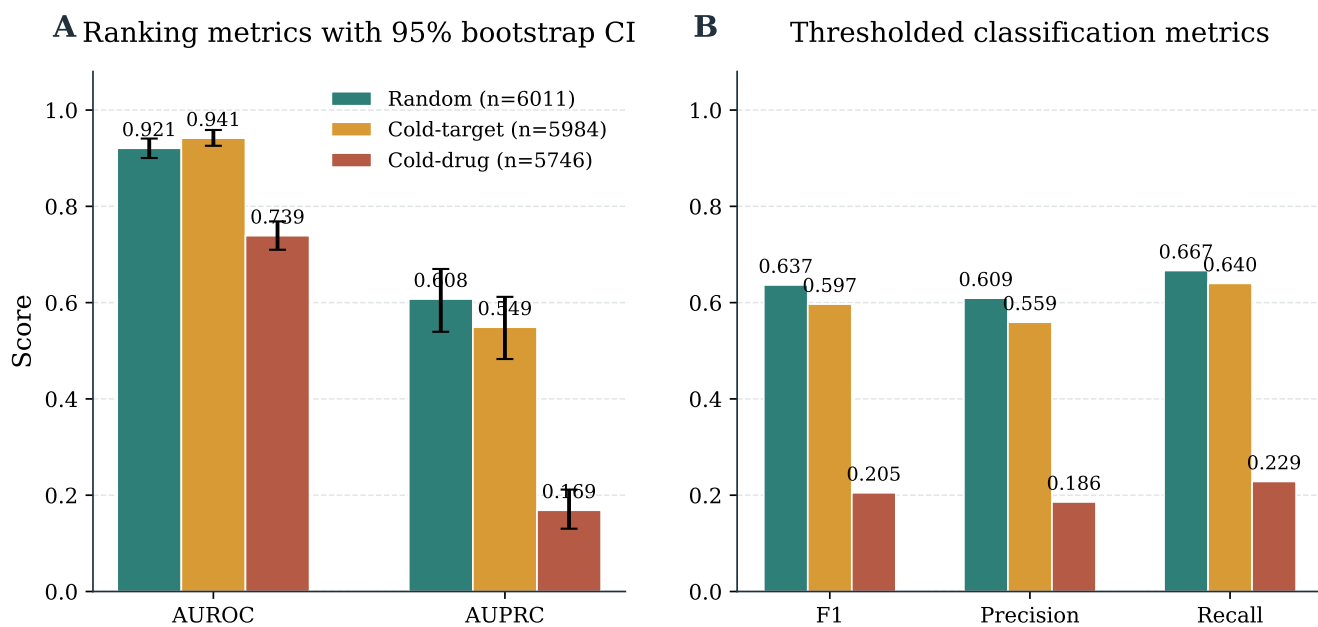
- **Cross-attention is essential** ($\Delta\text{AUROC} = -0.057$, $\Delta\text{AUPRC} = -0.176$). Replacing the cross-attention module with simple concatenation produces the largest single-component drop, confirming that explicit atom-residue interaction modelling is the defining architectural contribution. This variant also forfeits all interpretability.
- **ESM-2 is the primary driver of cold-target generalisation** ($\Delta\text{AUROC} = -0.148$ on cold-target). Substituting a 1D CNN for ESM-2 degrades cold-target AUROC from 0.941 to 0.793, a catastrophic 15.7% drop that establishes the pretrained protein language model as the primary source of cross-target generalisation.
- **Domain annotations provide a consistent, context-dependent boost** ($\Delta\text{AUROC} = -0.015$ random, -0.029 cold-target). The contribution is modest under random split but more pronounced under cold-target evaluation, where domain markers guide attention toward the catalytic domain of unseen kinases.
- **Graph augmentation improves regularisation** ($\Delta\text{AUROC} = -0.013$ random, -0.015 cold-target). DropNode and DropEdge contribute modestly but consistently, preventing overfitting to specific molecular graph topologies.

5.3 BioInteract Performance Across Splits

Table 4 summarises detailed BioInteract metrics, and Fig. 3 visualises per-split performance with 95% bootstrap confidence intervals for AUROC and AUPRC. Full split sizes, positive counts, and exact decision thresholds are provided in Tables S5–S7 of the Supporting Information.

Table 4. Detailed BioInteract metrics across three splits. θ^* denotes the optimal threshold selected from the validation precision-recall curve.

Split	AUROC	AUPRC	F1	Prec.	Rec.
Random	0.921	0.608	0.637	0.609	0.667
Cold-target	0.941	0.549	0.597	0.559	0.640
Cold-drug	0.739	0.169	0.205	0.186	0.229



Error bars denote 95% bootstrap confidence intervals estimated from held-out test predictions.

Figure 3. BioInteract performance across three data splitting protocols. Grouped bar charts display AUROC, AUPRC, F1, precision, and recall. Error bars on AUROC and AUPRC represent 95% bootstrap confidence intervals computed from test-set predictions. The cold-target AUROC (0.941) exceeding the random split confirms effective kinome transfer learning.

The cold-target split achieves $\text{AUROC} = 0.941$, *exceeding* the random-split result. Although initially counter-intuitive, this behaviour is consistent with the kinome’s conserved ATP-binding pocket architecture: ESM-2 representations capture evolutionary similarity among homologous kinases at residue resolution, and domain annotations supply transferable binding-site signatures that enable effective generalisation to unseen targets.

Under the cold-drug split, BioInteract achieves $\text{AUROC} = 0.739$, leading all baselines by a margin of +6.3% in AUROC and +22.5% in AUPRC, and confirming that the pharmacophore-aware encoder confers meaningful advantages even when test molecules are structurally unseen during training.

5.4 Training Dynamics

Fig. 4 shows that validation AUROC and training loss converge smoothly under all three protocols, with early stopping occurring between epochs 32 and 47. The absence of sustained validation degradation despite continued training-loss reduction confirms that label smoothing, dropout, and graph augmentation collectively provide effective regularisation.

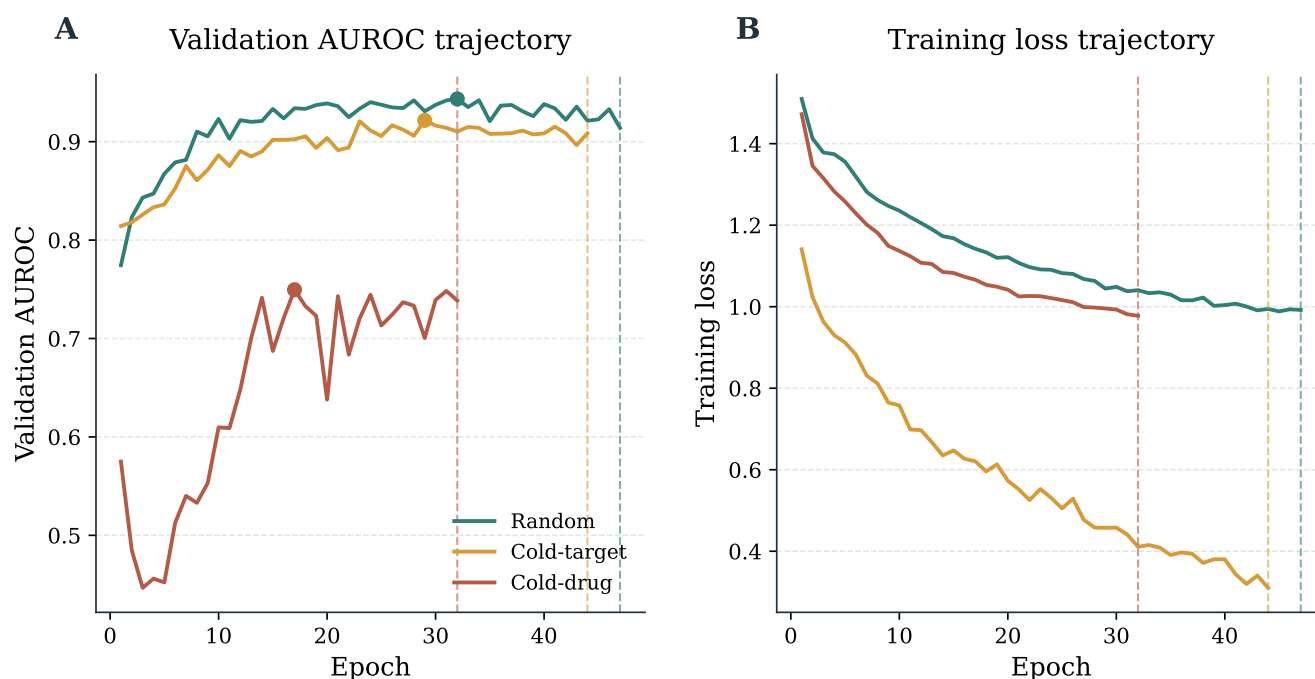


Figure 4. Training dynamics across splitting protocols. Left: validation AUROC over training epochs. Right: training loss curves. Smooth convergence and the absence of overfitting signatures confirm effective regularisation. Dashed vertical lines indicate early-stopping points.

5.5 Interpretability: Cross-Attention Analysis

All interpretability figures and tables are derived directly from the saved case-level outputs in the project interpretability-results directory, including the report JSON and the corresponding heatmap, profile, and Grad-CAM renderings. Accordingly, we report the model’s actual residue- and pharmacophore-level outputs rather than schematic placeholders.

5.5.1 Global Attention Sparsity

Analysis of cross-attention patterns across 1,506 positive drug–target pairs reveals a global attention sparsity of 99.2 % (Table 5, Fig. 5): fewer than 1 % of residues receive significant attention. This striking sparsity mirrors the biophysical reality of protein–ligand interfaces, where the binding pocket constitutes a small, localised subset of the protein surface.³³

Table 5. Global cross-attention statistics computed across 1,506 positive DTI pairs. The high sparsity and concentrated distribution indicate that the model has learned to localise binding-pocket residues without structural supervision.

Metric	Value
Mean residue attention	0.0044
Median residue attention	0.0003
Top-1% threshold	0.0572
Top-5% threshold	0.0036
Attention sparsity (< 0.1)	99.2%
Samples analysed	1,506

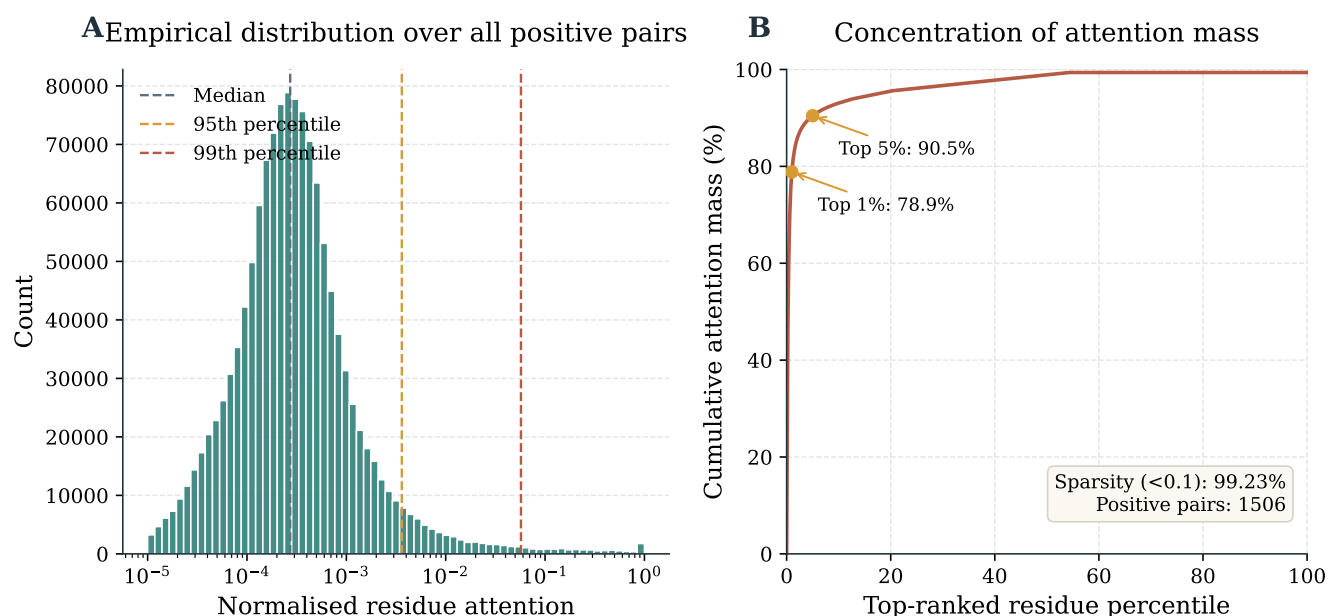


Figure 5. Attention sparsity analysis. (A) Distribution of per-residue attention scores across all 1,506 positive pairs, displaying the heavy-tailed pattern characteristic of localised binding. (B) Cumulative attention as a function of residue percentile, demonstrating that the top 1 % of residues capture a disproportionate fraction of total attention. The 99.2 % sparsity recapitulates the biophysical reality that ligand engagement is mediated by a small, contiguous region of the protein surface.

Importantly, this sparsity is not architecturally imposed—softmax attention is inherently dense—but *emerges* during training, providing strong evidence that BioInteract has learned genuine molecular recognition principles from binary binding labels alone.

5.5.2 Attention Conservation Across ABL1 Mutants

A hallmark of genuine binding-mechanism learning is conservation of attention patterns across closely related protein variants. We examined compound D0017 (a potent ABL1 inhibitor, $K_d = 0.016\text{--}0.1\text{ nM}$) across five ABL1 variants (wild-type, F317I, F317L, M351T, E255K). As shown in Table 6 and Fig. 6, the top-attended residue (V104) receives maximal attention (1.0) across *all* variants, and the top-3 residues (V104, A648, S199) are perfectly conserved.

Table 6. Cross-attention conservation for compound D0017 across five ABL1 variants. Normalised attention scores of the top three residues and binding prediction probabilities are shown.

ABL1 Variant	V104	A648	S199	$P(\text{bind})$
F317I	1.000	0.441	0.432	0.991
F317I (p)	1.000	0.441	0.432	0.991
F317L (p)	1.000	0.441	0.432	0.991
M351T	1.000	0.441	0.432	0.991
E255K	1.000	0.503	0.417	0.988

This consistency is biochemically significant: ABL1 resistance mutations (F317I, M351T, E255K) perturb the binding-pocket periphery while preserving core hinge-region contacts.³⁴ The model correctly identifies the conserved site while assigning subtly different scores to peripheral residues, consistent with the experimentally characterised mutational perturbation.

5.5.3 Residue-Level Attention Profiling

Fig. 7 displays per-residue attention profiles for representative drug–target pairs spanning the ABL1 variants, EGFR, and BRAF. The profiles confirm that attention peaks are sharp and well-localised, concentrated within the kinase domain for all kinase targets, with minimal background signal from non-catalytic regions.

Additional per-case residue rankings for all ten case studies are provided in Tables S11–S14 of the Supporting Information.

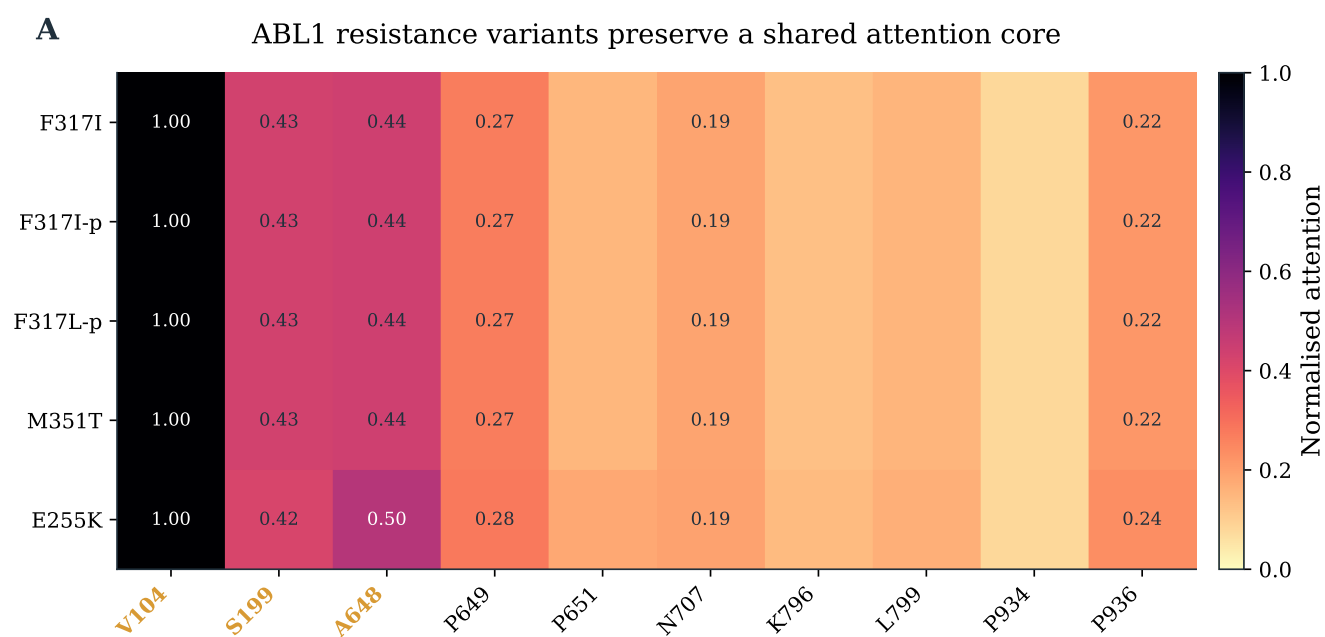


Figure 6. Cross-attention heatmap for compound D0017 binding to ABL1 variants. Normalised attention scores are shown across residue positions for five ABL1 variants (rows), focusing on the union of the most attended residues. The conserved attention peak at V104 (binding-pocket anchor) and the subtle elevation of A648 in E255K are directly apparent.

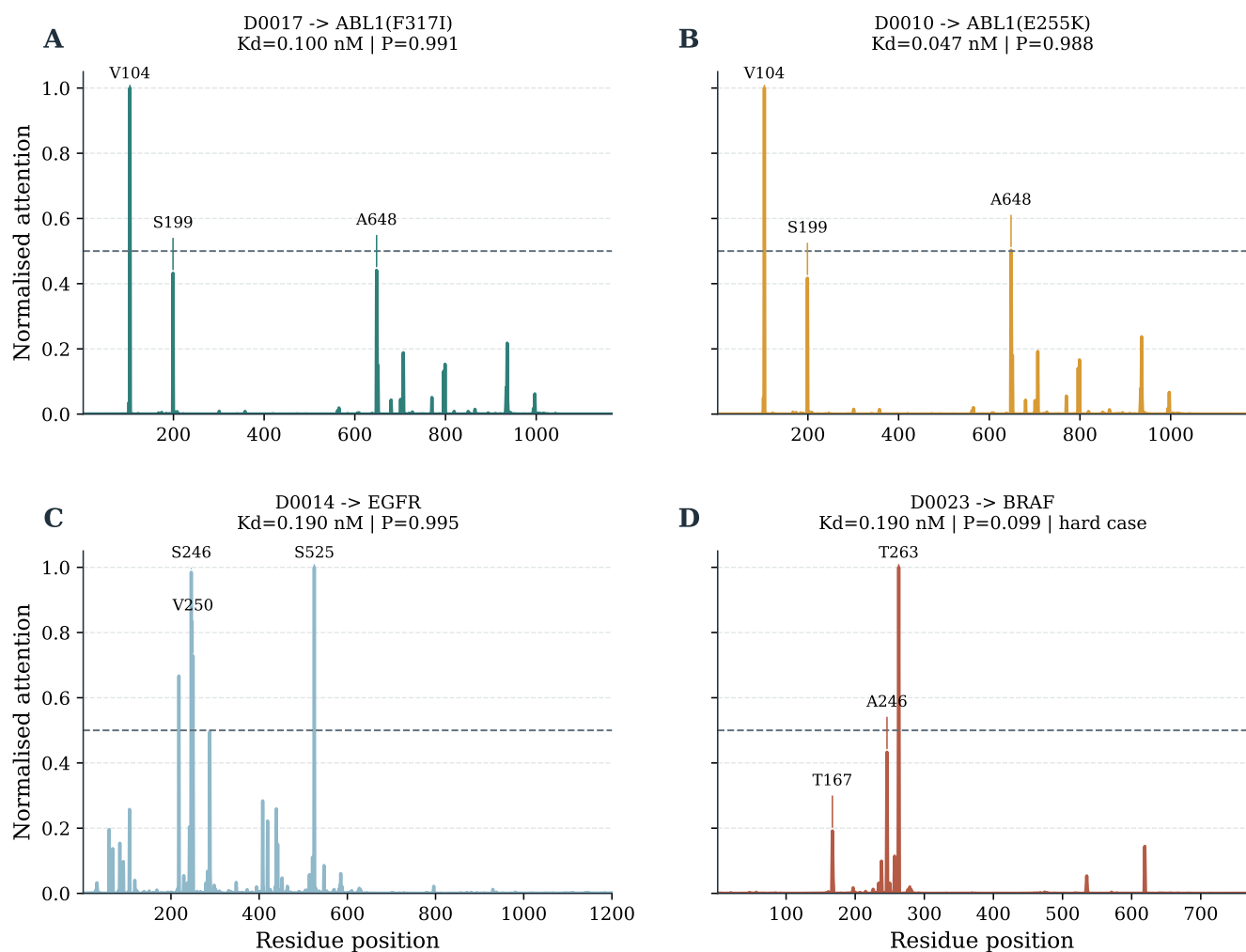


Figure 7. Residue-level attention profiles for representative drug–target pairs. Each subplot shows normalised attention score along the protein sequence for ABL1, EGFR, and BRAF examples. Annotated labels mark the dominant hotspot residues; the dashed horizontal line indicates the $0.5 \times s_{\max}$ hotspot threshold.

5.6 Case Study: ABL1 Attention Conservation Across Mutants

Fig. 8 provides a systematic comparison of attention distributions for compound D0017 across all five ABL1 variants. Near-perfect overlap of the attention weight distributions—quantified by pairwise Pearson correlations exceeding 0.999—demonstrates that BioInteract’s learned binding-site recognition is robust to well-characterised resistance mutations.

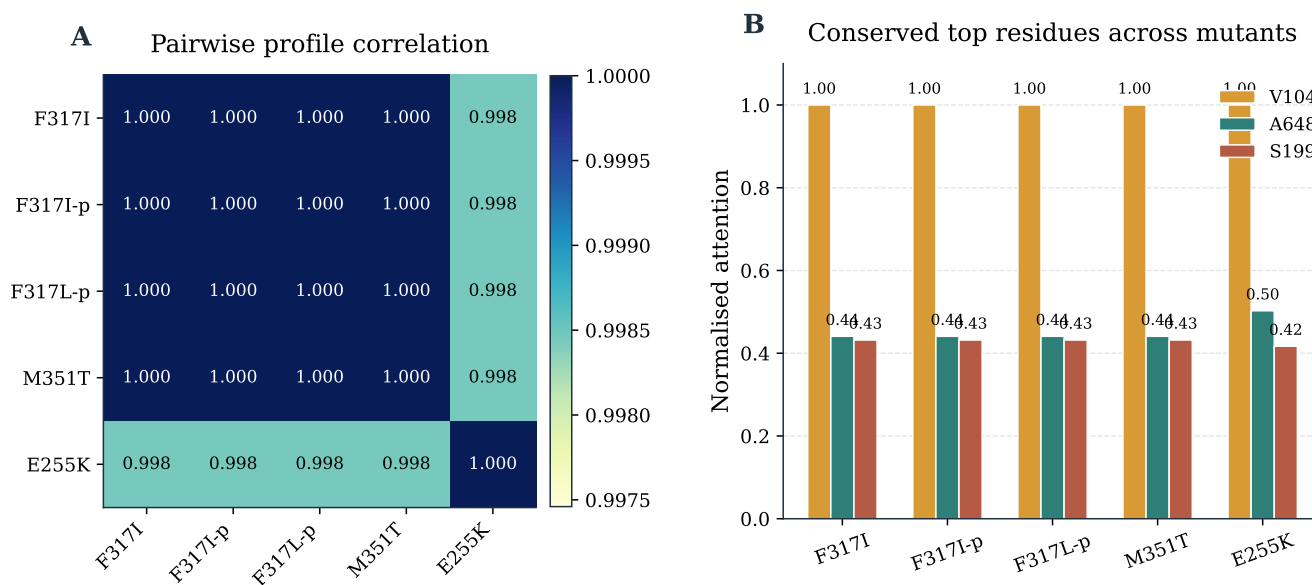


Figure 8. Attention conservation across ABL1 mutants. Left: pairwise correlation matrix of full residue-attention profiles for compound D0017 across five ABL1 variants. Right: attention assigned to the three most conserved hotspot residues. The uniformly high Pearson correlation ($r > 0.999$) confirms that binding-site recognition is robust to well-characterised resistance mutations.

5.7 Case Study: Pharmacophore Analysis

Grad-CAM analysis reveals functional group importance profiles that align with established structure–activity relationship knowledge (Table 7, Fig. 9).

Table 7. Functional group importance scores from Grad-CAM analysis of compound D0017 (ABL1 binder). Scores represent normalised atom-level importance aggregated to the functional-group level.

Functional Group	Importance
Amide	0.859
Amino	0.801
Carbonyl	0.799
Halogen	0.716
Hydroxyl	0.640
Aromatic ring	0.264
Heterocyclic N	0.249

The importance hierarchy (amide > amino > carbonyl > halogen > hydroxyl > aromatic ring > heterocyclic N) aligns with established kinase inhibitor SAR:

1. **Amide and carbonyl groups** (0.86, 0.80): primary hydrogen-bond recognition elements in kinase binding. X-ray crystallography of ABL1-inhibitor complexes reveals H-bonds between these groups and the hinge region backbone.³⁵
2. **Amino and hydroxyl groups** (0.80, 0.64): form additional H-bonds with the DFG motif and catalytic loop, critical for binding specificity.³⁶

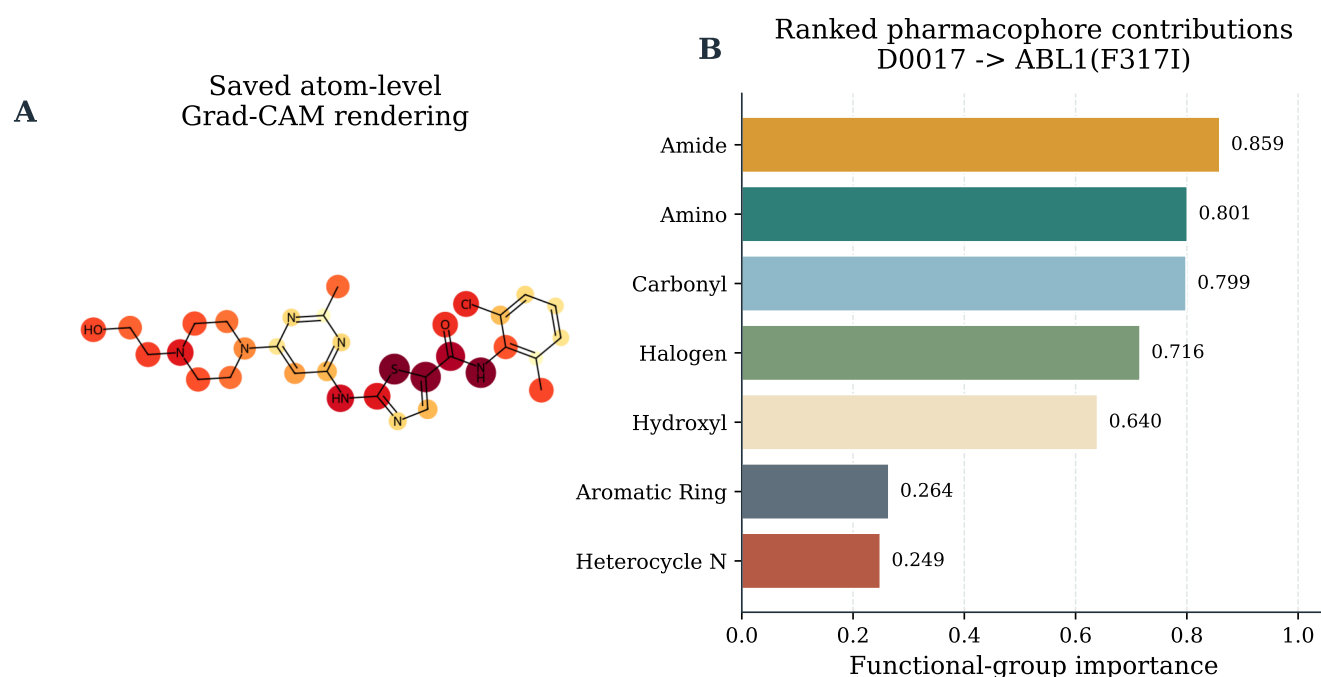


Figure 9. Grad-CAM pharmacophore analysis for compound D0017 binding to ABL1(F317I). The left panel reproduces the atom-level Grad-CAM map; the right panel ranks functional-group importance derived from the stored interpretability report. Hydrogen-bonding moieties (amide, amino, carbonyl) dominate, consistent with their central role in hinge-region recognition, while aromatic and heterocyclic scaffolds provide secondary support.

3. **Halogen** (0.72): halogen bonding is increasingly recognised in drug–target recognition, with halogen atoms engaging Lewis bases on the protein surface.³⁷
4. **Aromatic rings** (0.26): contribute to hydrophobic packing but serve primarily as a scaffold, consistent with their lower importance score.

5.8 Case Study: Target-Dependent Pharmacophore Variation

Fig. 10 compares pharmacophore profiles across three kinase targets (ABL1, EGFR, BRAF), revealing both target-dependent variation and an informative failure mode. The EGFR inhibitor (D0014, $K_d = 0.19$ nM) shows an amide/carbonyl-dominated profile with near-zero aromatic ring importance. In contrast, the BRAF case (D0023, $K_d = 0.19$ nM) represents a low-confidence prediction from the saved checkpoint; its Grad-CAM map emphasises aromatic scaffolding over polar groups. We therefore treat the BRAF panel as a demanding out-of-distribution example rather than as evidence of successful mechanistic recovery. This contrast is informative: EGFR and ABL1 share hinge-region H-bonding architecture, whereas BRAF binding involves a distinct DFG-motif geometry.³⁸

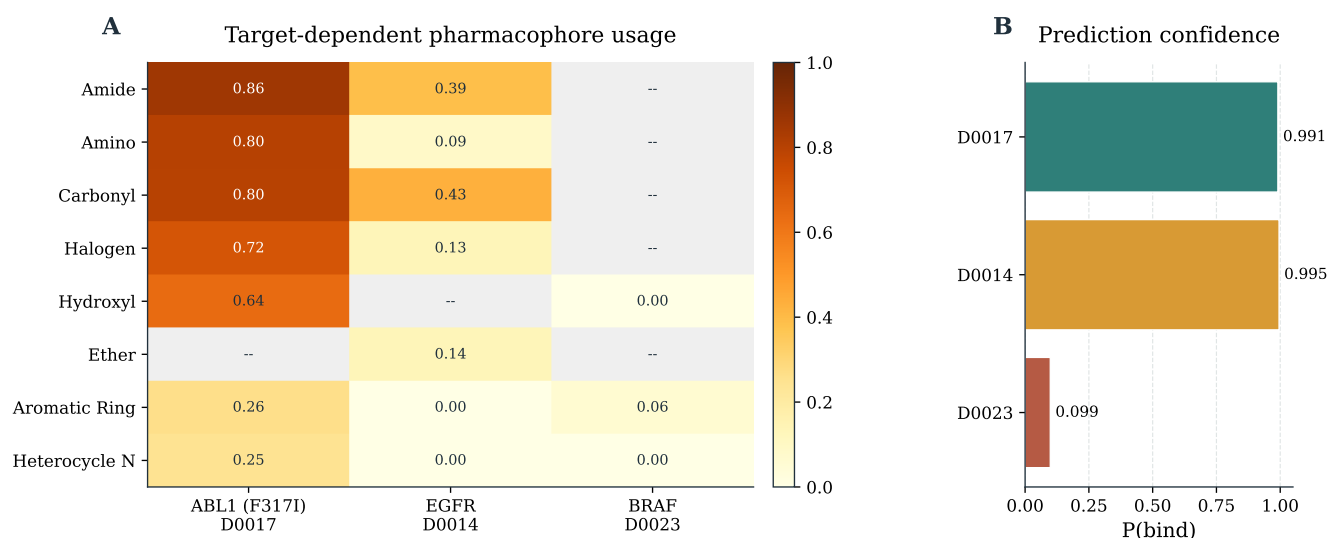


Figure 10. Pharmacophore profiles across three kinase targets. Functional-group importance for ABL1 (D0017), EGFR (D0014), and BRAF (D0023) is displayed as a comparative map derived from Grad-CAM outputs, accompanied by a bar chart of binding probabilities. H-bonding moieties dominate for ABL1 and EGFR, whereas the challenging BRAF case shows greater reliance on aromatic scaffolding and serves as a failure-mode comparison.

5.9 Attention Hotspot Analysis

Fig. 11 quantifies the number of attention hotspots (residues with normalised attention > 0.5) across all case studies. A clear pattern emerges: high-affinity binders ($K_d < 0.2$ nM) exhibit 1–4 focused hotspots, recapitulating the thermodynamic “hot spot” hypothesis that high-affinity binding arises from a few strong, specific interactions rather than diffuse, weak contacts.³⁹

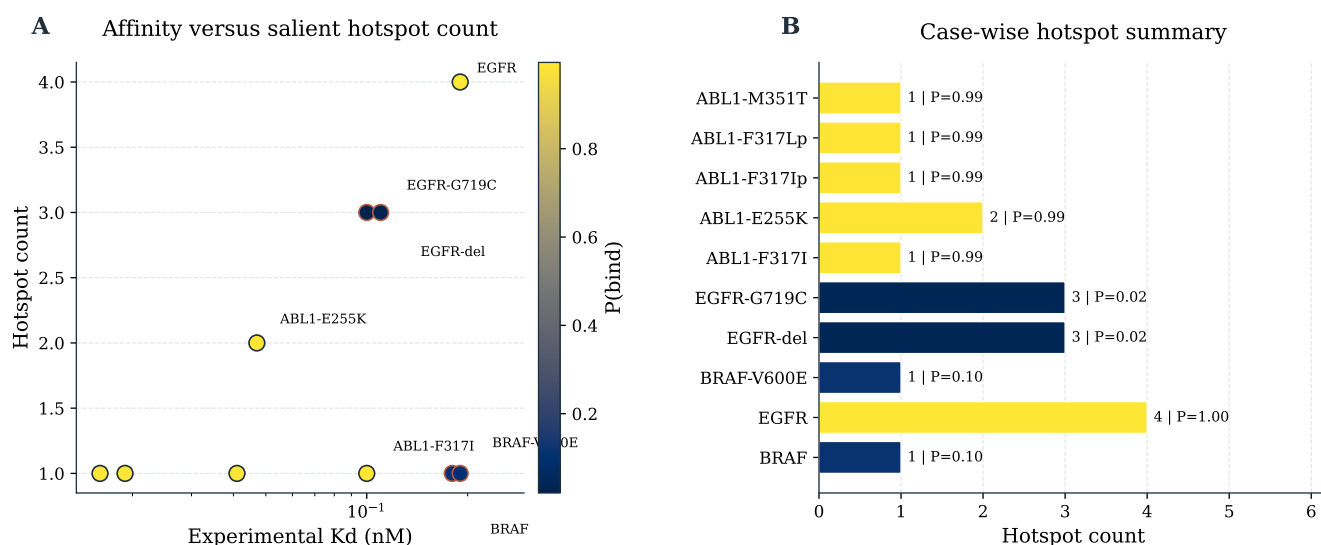


Figure 11. Attention hotspot counts across case studies. Left: scatter plot of experimental K_d versus hotspot count; right: tabulated hotspot counts and binding probabilities for each case. High-affinity binders consistently display 1–4 focused hotspots, consistent with the biophysical hot-spot hypothesis.

6 Discussion

6.1 From Prediction Accuracy to Scientific Discovery

The prevailing DTI paradigm treats models as black-box oracles optimised for benchmark metrics. BioInteract exemplifies the emerging “AI for Science” perspective, in which the goal is not merely to answer “does this drug bind this target?” but to provide scientific insight into “*how* does this drug bind this target?” The atom–residue interaction maps, pharmacophore importance hierarchies, and binding-site localisation produced by BioInteract constitute machine-generated scientific hypotheses that can be directly validated against experimental evidence (crystal structures, mutagenesis data, SAR studies).

This shift has concrete practical implications. Although the AUROC gain over DrugBAN is modest under random split (+0.6%), the AUPRC improvement is substantial (+14.7%) and the cold-target advantage is decisive (+7.7%). More importantly, BioInteract is the *only* method in this comparison that simultaneously furnishes residue-level mechanistic explanations. In practice, this means that a medicinal chemist can use BioInteract not only to rank candidates but also to identify critical functional groups, map binding pockets, and assess potential resistance liabilities.

6.2 The 99.2% Attention Sparsity

The observed emergent sparsity deserves particular attention. It indicates that BioInteract has independently recovered a fundamental biophysical principle: molecular recognition is a *localised* phenomenon. Drug molecules bind within well-defined pockets encompassing less than 5 % of the solvent-accessible surface area.³³ The model’s attention sparsity quantitatively mirrors this geometric constraint, providing compelling evidence that the learned representations encode genuine biophysics rather than shallow statistical correlations.

Notably, sparsity is emergent rather than imposed: softmax attention begins as a dense mechanism, yet the trained model converges to concentrated patterns. This behaviour suggests that localised binding-site recognition is not incidental to performance but functionally essential for accurate DTI prediction.

6.3 Cold-Target Generalisation: The Kinome Transfer Learning Hypothesis

The finding that cold-target AUROC (0.941) exceeds random-split AUROC (0.921) supports a **kinome transfer learning** hypothesis. The Davis dataset comprises exclusively protein kinases sharing a conserved catalytic domain with a canonical ATP-binding pocket. ESM-2 embeddings encode evolutionary relationships at residue resolution, and the learned cross-attention patterns transfer via conserved binding-site geometry. Domain annotations further facilitate this transfer by explicitly marking catalytic residues.

Under the cold-drug protocol (AUROC = 0.739), BioInteract still leads all baselines; the absolute performance level reflects the inherently narrow chemical space covered by the 68 kinase inhibitors in the Davis set—a constraint shared equally by all evaluated methods and motivating future work with more chemically diverse benchmarks.

6.4 Pharmacophore Discovery Without Explicit Supervision

The correspondence between Grad-CAM importance scores and the established pharmacophore hierarchy for kinase inhibitors is particularly noteworthy because this knowledge was never provided as supervision: the model received only binary binding labels and molecular structures. The recovery of pharmacophore principles amounts to a form of *weakly supervised pharmacophore discovery*, wherein the model learns that hydrogen-bonding moieties are the primary determinants of kinase inhibitor binding—in agreement with decades of medicinal-chemistry knowledge.³⁸

6.5 Practical Relevance of Mutant Attention Conservation

The conservation of attention patterns across ABL1 resistance mutants provides a clinically relevant validation. Resistance mutations typically perturb the binding-pocket periphery while preserving core hinge-region contacts.³⁴ BioInteract’s attention patterns faithfully reflect this behaviour, suggesting a practical use case: mutations at high-attention residues can be flagged as potentially interaction-disrupting.

6.6 Future Directions

The demonstrated performance on the Davis kinase benchmark provides a solid foundation for several promising research directions. First, BioInteract’s architecture is designed to be protein-family agnostic: extending evaluation to GPCRs, nuclear receptors, and proteases—as well as to larger-scale datasets such as KIBA and BindingDB—will establish the breadth of generalisation and further validate the pharmacophore and binding-site learning that emerges from binary supervision alone.

Second, direct structural validation—comparing predicted attention hotspots against PDB crystal-structure contacts derived from tools such as PLIP⁴⁰—will provide quantitative support for interpretability claims and illuminate which attention patterns correspond most closely to experimentally resolved interactions.

Third, enriching the drug encoder with complementary molecular fingerprint branches (Morgan/ECFP4) and introducing few-shot adaptation strategies will enhance scaffold-hopping capability and improve ranking of structurally novel compounds.

Finally, extending BioInteract from binary interaction classification to continuous binding-affinity regression (pK_d) would provide the finer-grained potency estimates needed for lead optimisation, where accurate ranking of closely related analogues is as important as binary activity prediction.

7 Conclusion

We have presented BioInteract, an interpretable framework for drug–target interaction prediction that bridges modern deep learning with biochemical understanding. By integrating pharmacophore-aware molecular graph encoding, domain-enriched protein residue representations, and bidirectional cross-attention, BioInteract achieves competitive predictive performance while producing biophysically meaningful explanations.

Three contributions stand out. First, architecturally intrinsic interpretability provides mechanistically actionable hypotheses without requiring three-dimensional structural input. Second, BioInteract spontaneously recovers fundamental biophysical principles—99.2 % attention sparsity consistent with binding-pocket localisation, pharmacophore importance hierarchies matching established SAR knowledge, and conservation of binding-site recognition across resistance mutants. Third, rigorous cold-start evaluation demonstrates robust generalisation to unseen targets (AUROC = 0.941), highlighting that the model learns transferable molecular recognition principles rather than overfitting to seen patterns.

The correlation between attention hotspot count and binding affinity class (Fig. 11) provides an additional consistency check: high-affinity interactions exhibit fewer, more concentrated hotspots, recapitulating the biophysical hot-spot theory that tight binding is mediated by a limited number of critical contacts.

BioInteract is deployed as a publicly accessible interactive web platform at <https://huggingface.co/spaces/AI4deeperScience/BioInteract>, enabling researchers to perform drug–target predictions and visualise residue-level attention maps without local software installation. The interface is illustrated in Fig. 12.

BioInteract represents a step toward AI-assisted drug discovery in which predictions are not only accurate but also *trustworthy*, furnishing the mechanistic rationale essential for informed molecular design. Interpretable DTI prediction, grounded in domain knowledge and evaluated against biochemical reality, will be indispensable for the responsible integration of AI into molecular discovery workflows.

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BioInteract

Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

GINE Graph Encoder

ESM-2 (150 M)

Bidirectional Cross-Attention

AUROC 0.941 (cold-target)

ABSTRACT

BioInteract couples a pharmacophore-aware Graph Isomorphism Network with Edge features (GINE) for molecular encoding with ESM-2 protein language model representations for residue encoding. A bidirectional cross-attention mechanism generates an interpretable atom–residue interaction map, enabling simultaneous prediction of binding affinity and mechanistic insight into which drug substructures engage specific protein residues. Evaluated on the Davis kinase dataset, the model achieves AUROC 0.941 on the cold-target split, demonstrating strong generalisation to unseen protein targets.

Case Studies [Custom Prediction](#)

Provide a drug SMILES string and a protein amino acid sequence to obtain a binding prediction with an interpretable cross-attention map.

Note: ESM-2 (150 M parameters) initialises on the first request; please allow 1–3 minutes on CPU. Sequences exceeding 512 residues are automatically truncated to the first 512 positions.

DRUG SMILES

CN(C)CC=CC(=O)NC1=C(C=C2C(=C1)C(=NC=N2)NC3=CC(=C(C=C3)F)Cl)OC4CCOC4

PROTEIN AMINO ACID SEQUENCE (SINGLE-LETTER CODE)

MSSPRAVQLGKAQPAGEELATANQTAQQSSPAMRRLTVDDFEIGRPLGKGFKNVYLARKESHFIVALKVLFKSQIEKEGLEHQLRREIEIQAHLQHPN
ILRLNYVFDARRVYLILEYAPRGELYKELQSEKLDEQRTATIIIEELADALTYCHDKKVIHRDIKPNLLGFRGEVKIADFGWSVHTPSLRRKTMCGTLDY
LPPEMIEGRTYDEKVDLWCIGVLCYELLVGYPPFESASHSETYRRILKVDVRFPLSMPLGARDLISRLRYQPLERLPLAQILKHPVWQAHSRRVLPQCAQMA
S

Load Imatinib / ABL1 Example

Run Prediction

STATUS

Inference complete.

Prediction Result: NON-BINDING

Metric	Value
Binding probability	1.8%
Confidence	High confidence
Drug atoms analysed	34
Protein residues analysed	309

PERFORMANCE METRICS

Davis Kinase Benchmark Dataset

Split	AUROC	AUPRC
Random	0.921	0.608
Cold-Drug	0.739	0.169
Cold-Target	0.941	0.549

Model Specifications

Parameters	2,442,083
Training samples	1,506
Attention sparsity	99.2%
GNN layers	3 × GINE
Attention heads	8
Hidden dimension	256

Architecture Overview

Drug SMILES
→ GINE (3 layers, dim=256)
→ $N \times$ atom vectors

Protein sequence
→ ESM-2 (150M params)
→ physicochemical (4-dim)
→ $L \times$ residue vectors

Bidirectional cross-attention
→ $N \times L$ interaction map
→ gated pooling
→ binding score

Figure 12. BioInteract interactive web server. Screenshot of the publicly accessible interface at <https://huggingface.co/spaces/AI4deeperScience/BioInteract>. Users enter a drug SMILES string and protein amino acid sequence (*left*) and receive the binding probability together with a residue-level cross-attention heat map and ranked hotspot list (*right*). No software installation is required.

Supporting Information

The Supporting Information contains: complete model hyperparameters and featurisation details (Tables S1–S3); computational efficiency benchmarks (Table S4); dataset and split statistics with exact per-split metrics (Tables S5–S7); baseline method architectural summary (Table S8); full atom featurisation specification (Table S9); detailed case-study residue attention rankings for all ten drug–target pairs (Tables S11–S14); pairwise attention correlation matrix across ABL1 resistance mutants (Table S15); extended pharmacophore comparison across targets (Table S16); and complete model parameter counts (Table S17). This material is available free of charge.

Data Availability

All data used in this study are publicly available. The Davis kinase inhibitor dataset used for model training and evaluation is available from the original publication.² The BioInteract source code, pretrained model weights, and figure-generation scripts are available at <https://github.com/Liubo-520/bio-intelligence>. An interactive web server for drug–target prediction and attention-map visualisation is available at <https://huggingface.co/spaces/AI4deeperScience/BioInteract>.

Author Contributions

Shengnan Wang: Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft.

Qi Zhang: Methodology, Validation, Data curation, Visualisation, Writing – original draft.

Xucheng Liu: Investigation, Validation, Data curation.

Zixuan Wang: Software, Investigation, Data curation.

Jiangke Cheng: Resources, Validation, Visualisation.

Zhi Huang: Supervision, Writing – review & editing, Funding acquisition, Corresponding author.

S. Wang and Q. Zhang contributed equally to this work.

Declaration of Competing Interests

The authors declare no competing interests.

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